

Adelowo Adebajji Oluwatimileyin

Applied Mathematician · Scientific Machine Learning Researcher

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PROFILE

M.Sc. Mathematical Engineer working at the intersection of numerical PDEs, computational fluid dynamics, inverse problems, and scientific machine learning. Experience developing and verifying spectral and finite-element solvers, reduced-order models, PDE-constrained inverse and optimisation methods, and Bayesian/data-assimilation workflows, with numerical verification through manufactured and exact solutions, convergence studies, and independent benchmarks. Complemented by industry experience building machine-learning models, signal-processing pipelines, data systems, and deployed REST APIs in Python, PyTorch, NumPy/SciPy, SQL, and FastAPI.

RESEARCH INTERESTS

Numerical PDEs and scientific computing; computational fluid dynamics; inverse problems and PDE-constrained optimisation; reduced-order modelling and operator learning; scientific machine learning; dynamical systems and data assimilation; uncertainty quantification.

PROFESSIONAL EXPERIENCE

Dec 2024 – Present	Data Scientist <i>Intuos Srl</i> , L'Aquila, Italy - Developed flight-phase classifiers served through a query-on-demand dashboard for recorded aircraft telemetry. - Designed FFT-based signal-processing pipelines for aircraft engine RPM estimation, frequency analysis, and anomaly detection. - Built operational analytics workflows over externally ingested telemetry for flight-performance monitoring. - Delivered REST APIs exposing machine learning models and engineering analytics to operational systems.
Apr 2023 – Nov 2024	Machine Learning Engineer <i>N-and Italia Srl</i> , Italy - Developed predictive machine learning models for vending-machine delivery-failure forecasting using large-scale transaction and telemetry data. - Performed exploratory and statistical analysis to identify patterns affecting system performance and operational reliability. - Built data-processing and KPI-monitoring workflows to support data-driven operational decisions.
Jan 2023 – Mar 2023	Operational Data Analyst <i>eResult Srl</i> , Cesena, Italy - Optimised SQL queries and automated recurring reporting workflows for business-intelligence systems. - Developed reproducible data-processing workflows and transformed operational data into actionable performance metrics.
Apr 2022 – Oct 2022	Digital Engineering Intern <i>Bridgestone EMIA</i> , Rome, Italy - Developed computational-geometry algorithms for engineering simulation and finite-element meshing workflows. - Improved data-structure operations for tire cross-section geometry processing and implemented closest-point algorithms for cubic Bézier curves.
May 2022 – Aug 2022	Machine Learning Intern <i>Fellowship.AI</i> , Remote - Implemented an InsetGAN architecture for whole-body human-image generation and developed the associated deep-learning inference workflow. - Deployed model inference through FastAPI for integration with a Swift iOS application.

SELECTED PROJECTS

Applied Mathematics & Scientific Computing

2D Incompressible Navier–Stokes: Pseudo-Spectral Solver and POD-Galerkin ROM

[GitHub repository](#)

- Developed and verified a Fourier pseudo-spectral solver for 2D incompressible Navier–Stokes with integrating-factor RK4 and 2/3-rule dealiasing, confirming fourth-order temporal and spectral spatial convergence against the Taylor–Green exact solution and an independent manufactured solution.
- Derived a tensorial POD-Galerkin reduced-order model with DEIM hyper-reduction, verified to machine precision against the projected full-order operators; achieved $38.5\times$ online speed-up at $<0.1\%$ state error at rank 16, and quantified a snapshot-POD failure to generalise across unseen flow realisations ($\approx 99\%$ error).

Finite-Element Solver for 2D Incompressible Flow: Schäfer–Turek Cylinder Benchmarks

[GitHub repository](#)

- Implemented a Taylor–Hood P_2/P_1 FEniCSx/PETSc solver for incompressible viscous flow, verified by manufactured solutions recovering the theoretical $O(h^3)$ velocity- L^2 and $O(h^2)$ velocity- H^1 /pressure- L^2 convergence rates.
- Verified the solver on the DFG Schäfer–Turek cylinder benchmarks, adding degree-2 curved geometry, audited force definitions and periodic-cycle extraction; the finest steady 2D-1 result closely matches the published spectral reference.
- In a five-level unsteady 2D-2 study (to 273k DOFs, with spatial and temporal refinement), drag and Strouhal number lie within their inherited benchmark ranges and the pressure difference is within range but not shown to be spatially converged; the refined lift maximum (≈ 0.987) is slightly below the inherited 0.990 lower bound while closely matching a published FeatFlow computation, so full reproduction is not claimed.

Deterministic and Bayesian Inversion of Spatially Varying Darcy Permeability

[GitHub repository](#)

- Developed adjoint-based deterministic inversion for a spatially varying log-permeability field (UFL automatic differentiation, H^1/L^2 Tikhonov regularisation with discrepancy-principle selection, L-BFGS-B), verified by a Taylor-remainder test, and diagnosed a regularisation-model mismatch on structured (inclusion-like) truth fields.
- Extended to Bayesian inversion with a KL-parameterised Gaussian prior sampled by preconditioned Crank–Nicolson MCMC, independently validated against an analytical linear-Gaussian posterior (0.59% error); reported MAP rather than the posterior mean as the primary estimate given diagnosed low effective sample size.

Phase-Field Modelling of Boiling Heat Transfer: Toward 3D DNS

[GitHub repository](#)

- Built a conservative Allen–Cahn/Navier–Stokes/energy phase-field solver for vapour-liquid phase change (Roccon 2025), exploiting a constant-coefficient FFT pressure-Poisson solve that extends unchanged to 3D; the 2D bubble-growth benchmark agrees with the analytical growth rate, with the error decreasing systematically under grid refinement.
- Diagnosed the failure of an initial 1D Stefan-problem implementation and developed a dedicated non-periodic 1D Stefan solver that converges at first order (interface-position error 1.142% to 0.079% for $\Delta x = 2.0\text{--}0.125$ mm at $t = 250$ s, matched densities) with verified mass and energy balances; numerically verified the 3D solver (Poisson/projection convergence, phase-mass identity, 2D/3D consistency; this exposed and corrected a missing 3D viscous term). The general multiphase heat-flux pathway remains under validation and 3D physical validation is not yet done.

2D Elasticity FEM and SIMP Topology Optimisation

[GitHub repository](#)

- Verified a P_1 plane-stress elasticity solver by the method of manufactured solutions and a SIMP compliance sensitivity by a Taylor-remainder test before trusting it for optimisation; reduced canonical cantilever compliance from 617.86 to 101.39 under a volume constraint with Bourdin density filtering.
- Characterised the method's limits directly: the mesh-resolution study is mesh-independent-looking but not quantitatively mesh-converged ($\approx 14\%$ compliance gap between the two finest meshes), and the optimality-criteria convergence metric does not decrease monotonically.

Ensemble Kalman Filtering for the Chaotic Lorenz-96 System

[GitHub repository](#)

- Implemented and independently validated a stochastic Ensemble Kalman Filter (perturbed observations) for the chaotic Lorenz-96 system ($K = 40$, $F = 8$) against a linear/Gaussian Kalman-filter problem; estimated the largest Lyapunov exponent $\lambda_1 \approx 1.69$ via Benettin renormalisation.
- Diagnosed severe ensemble under-dispersion and filter collapse at small ensemble sizes, and reduced assimilation RMSE by over $11\times$ using Gaspari–Cohn covariance localisation, quantifying ensemble-size, inflation, and observation-density/noise effects.

Applied ML & Engineering

INTUOS Flight Data Monitoring Dashboard: Full-stack aviation analytics platform (FastAPI + IBM DB2, React/Vite, Docker) serving ML-based flight-phase classifiers over recorded telemetry (weighted F1 0.999 against labels derived from an existing model on row-level splits; held-out-flight evaluation is future work); developed at Intuos Srl.

LLM Engineering Portfolio: Nine applied machine-learning and generative-AI projects, including a RAG assistant, an NL-to-SQL pipeline, and a FinBERT compression pipeline (head pruning, layer dropping and knowledge distillation, 110M to 88M parameters) whose test split likely overlaps the baseline's fine-tuning data.

Full portfolio, including additional applied-math and engineering projects: adebanjioluwatimileyin.github.io/projects.html

TECHNICAL SKILLS

Mathematical / Numerical: Numerical PDEs, finite-element methods, pseudo-spectral methods, variational methods, reduced-order modelling (POD–Galerkin), hyper-reduction (DEIM), inverse problems, PDE-constrained optimisation, Tikhonov/variational regularisation, Bayesian inference (MCMC/pCN), ensemble data assimilation

Scientific ML / ML: PyTorch, TensorFlow, scikit-learn, physics-informed neural networks, Fourier Neural Operators, surrogate/operator learning, supervised learning, model evaluation

Programming / Scientific Computing: Python, NumPy, SciPy, MATLAB, C++, FEniCSx, PETSc, JAX, FFT-based computing

Engineering / Data: SQL, IBM DB2, PostgreSQL, FastAPI, Docker, Git, REST APIs, ETL/data pipelines, AWS

EDUCATION

2018 – 2021	M.Sc. Mathematical Engineering University of L'Aquila, Italy Final grade: 106/110 <i>Thesis:</i> “Bounds on mixing of passive scalars advected by incompressible energy and enstrophy constrained flows and Anomalous dissipation” (code and report) – studied lower bounds on the mix norm (Iyer–Kiselev–Xu) and optimal stirring (Lin–Thiffeault–Doering) for passive scalars in incompressible, enstrophy-constrained flow, simulating optimal stirring with a pseudo-spectral solver; later extended by a resolution study to $N = 512$. <i>Supervisor:</i> Prof. Stefano Spirito. <i>Relevant coursework:</i> Functional Analysis, Applied PDEs, Fluid Dynamics, Numerical Methods for PDEs, Finite Element Methods, Scientific Computing, Dynamical Systems and Bifurcation Theory, Advanced Optimisation.
2014 – 2018	B.Sc. Mathematics Obafemi Awolowo University, Nigeria First Class Honours <i>Thesis:</i> Application of Galerkin Finite Element Method to the One-Dimensional Stefan Problem.

RESEARCH EXPERIENCE

2020 – 2021	M.Sc. Thesis Research University of L'Aquila Studied analytical lower bounds on passive-scalar mixing in incompressible, enstrophy-constrained flows and simulated Lin–Thiffeault–Doering optimal stirring with a pseudo-spectral Python solver, comparing fitted mixing rates with the Iyer–Kiselev–Xu scaling.
2017 – 2018	B.Sc. Thesis Research Obafemi Awolowo University Applied the Galerkin finite-element method to the one-dimensional Stefan free-boundary problem, developing numerical algorithms for phase-boundary tracking.

AWARDS & LEADERSHIP

Awards

University of L'Aquila International Scholarship	2018 – 2021
Professor S.S. Okoya Prize for Excellence in Applied Mathematics	2018
MTN Foundation Scholarship	2014 – 2018
PTDF Scholarship	2014 – 2018
Murli T. Chellaram Foundation Scholarship	2014 – 2018

Teaching & Leadership

2016 – 2018	Founder & President, Inter-University Mathematics Club Nigeria Established a cross-university mathematics mentorship network and organised seminars on mathematics, research, and academic development.
2015 – 2018	Academic Director, NAMSS Peer Tutor Obafemi Awolowo University Coordinated workshops and tutored undergraduate students in mathematical analysis and quantitative reasoning.

REFERENCES

Prof. Ogundare Babatunde Sunday Professor, Obafemi Awolowo University, Nigeria bogunda@oauife.edu.ng	Dr. Yusuf Tijani Lecturer, University of the Witwatersrand, South Africa yusuf.tijani@wits.ac.za
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